

Math 142, S12

Quiz 8

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Determine whether the series is convergent or divergent using the Integral Test.

$$(a) \sum_{n=1}^{+\infty} \frac{1}{(n+1)^3}$$

$$f(x) = \frac{1}{(x+1)^3}$$

$$\begin{aligned} \int_1^{+\infty} \frac{1}{(x+1)^3} dx &= \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{(x+1)^3} dx \\ &= \lim_{t \rightarrow +\infty} \left[-\frac{1}{2(x+1)^2} \right]_1^t \\ &= \lim_{t \rightarrow +\infty} \left[-\frac{1}{2(t+1)^2} + \frac{1}{2 \cdot 2^2} \right] \\ &= \frac{1}{8} \end{aligned}$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{1}{(n+1)^3}$ is convergent

$$(b) \sum_{n=2}^{+\infty} \frac{1}{n \ln n}$$

$$f(x) = \frac{1}{x \ln x}$$

$$\begin{aligned} \int_2^{+\infty} \frac{1}{x \ln x} dx &= \lim_{t \rightarrow +\infty} \int_2^t \frac{1}{x \ln x} dx \\ &= \lim_{t \rightarrow +\infty} \ln(\ln x) \Big|_2^t \\ &= \lim_{t \rightarrow +\infty} \ln(\ln t) - \ln(\ln 2) \end{aligned}$$

$$= +\infty$$

$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n \ln n}$ is divergent.

Problem 2. (5 points) Determine whether the series is convergent or divergent using the Comparison Test or the Limit Comparison Test.

$$(a) \sum_{n=1}^{+\infty} \frac{\cos^2 n}{n^2}$$

$$a_n = \frac{\cos^2 n}{n^2}$$

$$\text{Let } b_n = \frac{1}{n^2}$$

clearly $\cos^2 n \leq 1$, then

$$\frac{\cos^2 n}{n^2} \leq \frac{1}{n^2}$$

Since $\sum_{n=1}^{+\infty} \frac{1}{n^2}$ is convergent, we

obtain $\sum_{n=1}^{+\infty} \frac{\cos^2 n}{n^2}$ is convergent.

$$(b) \sum_{n=1}^{+\infty} \frac{n+1}{2n^3+1}$$

$$a_n = \frac{n+1}{2n^3+1}, \text{ let } b_n = \frac{n}{2n^3} = \frac{1}{2n^2}$$

and $\sum_{n=1}^{+\infty} b_n = \frac{1}{2} \sum_{n=1}^{+\infty} \frac{1}{n^2}$ is convergent

$$\begin{aligned} \lim_{n \rightarrow +\infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow +\infty} \frac{\frac{n+1}{2n^3+1}}{\frac{1}{2n^2}} = \lim_{n \rightarrow +\infty} \frac{n+1}{2n^3+1} \cdot 2n^2 \\ &= \lim_{n \rightarrow +\infty} \frac{2n^3+2n^2}{2n^3+1} \cdot \frac{1}{n^3} \\ &= \lim_{n \rightarrow +\infty} \frac{2+\frac{2}{n}}{2+\frac{1}{n^3}} = \frac{2}{2} = 1 \end{aligned}$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{n+1}{2n^3+1}$ is convergent.

(You may choose $b_n = \frac{1}{n^2}$ too).